

BirTics Final Wiring

1. ESP32 GPIO Map

Device	Signal	ESP32 GPIO
L298N	ENB	26
L298N	IN3	14
L298N	IN4	27
I2C	SDA	21
I2C	SCL	22
Servo	Signal	18
ToF Left	XSHUT	32
ToF Right	XSHUT	33
ToF Front	XSHUT	25
Color Sensors	S0	4
Color Sensors	S1	16
Color Sensors	S2	17
Color Sensors	S3	5
Ground Color	OUT	34
Front Color	OUT	35

2. I2C Expansion Board

Expansion Port	Connected To	Pins
Input	ESP32	3.3V, GND, SDA21, SCL22
Port 1	MPU6050	VCC, GND, SDA, SCL
Port 2	ToF Left	VIN, GND, SDA, SCL
Port 3	ToF Right	VIN, GND, SDA, SCL
Port 4	ToF Front	VIN, GND, SDA, SCL

3. Color Sensors

Pin	Ground Sensor	Front Sensor
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VCC	5V	5V
GND	GND	GND
LED	5V	5V
S0	GPIO4	GPIO4
S1	GPIO16	GPIO16
S2	GPIO17	GPIO17
S3	GPIO5	GPIO5
OUT	GPIO34	GPIO35

4. Power

Part	Power	Notes
Servo	5V + GND	Signal GPIO18
L298N (Channel B)	Battery	ENB, IN3, IN4
ESP32	USB / Shield	Common GND
ToF + MPU	3.3V via I2C board	
Color Sensors	5V	LED on 5V